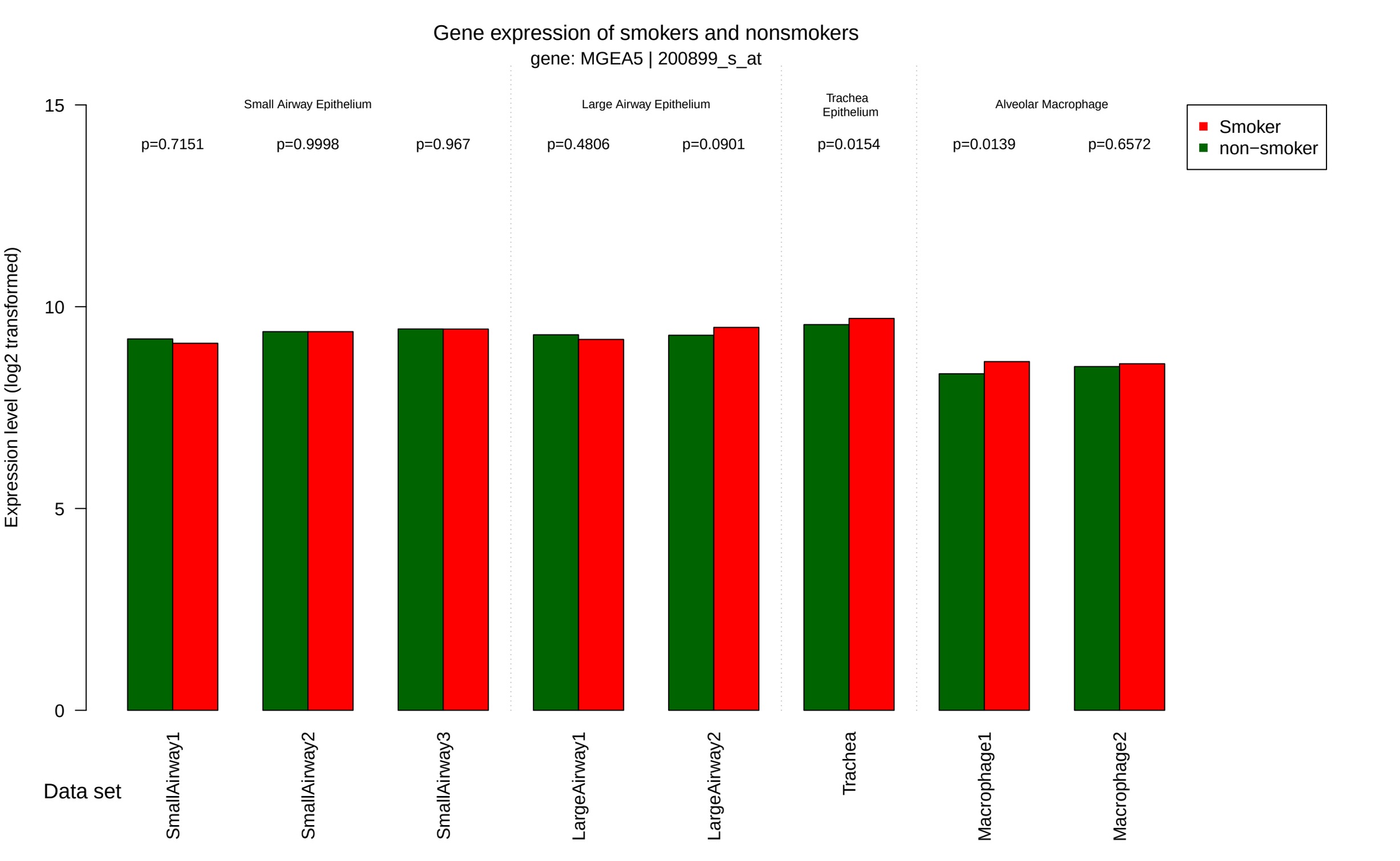
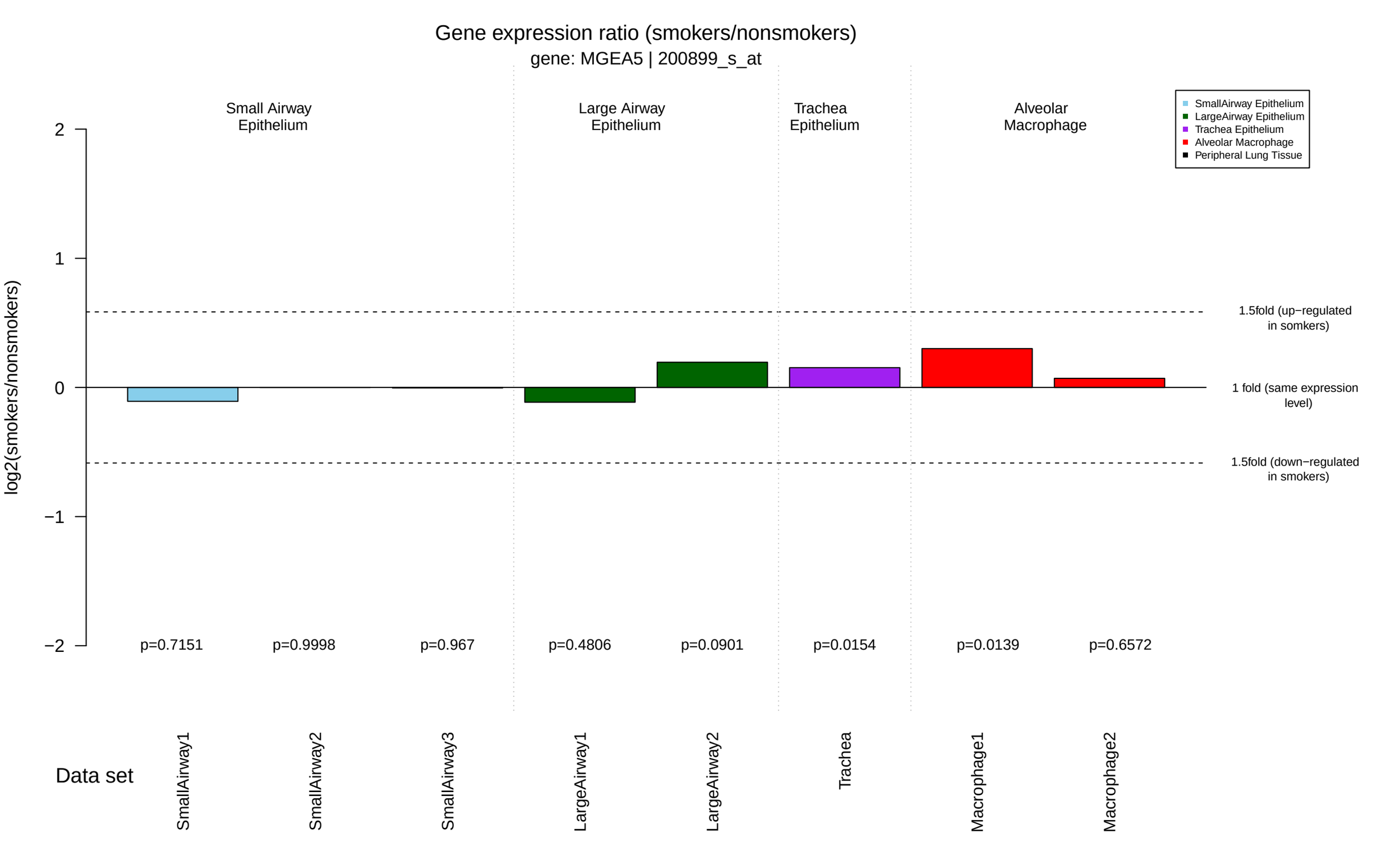
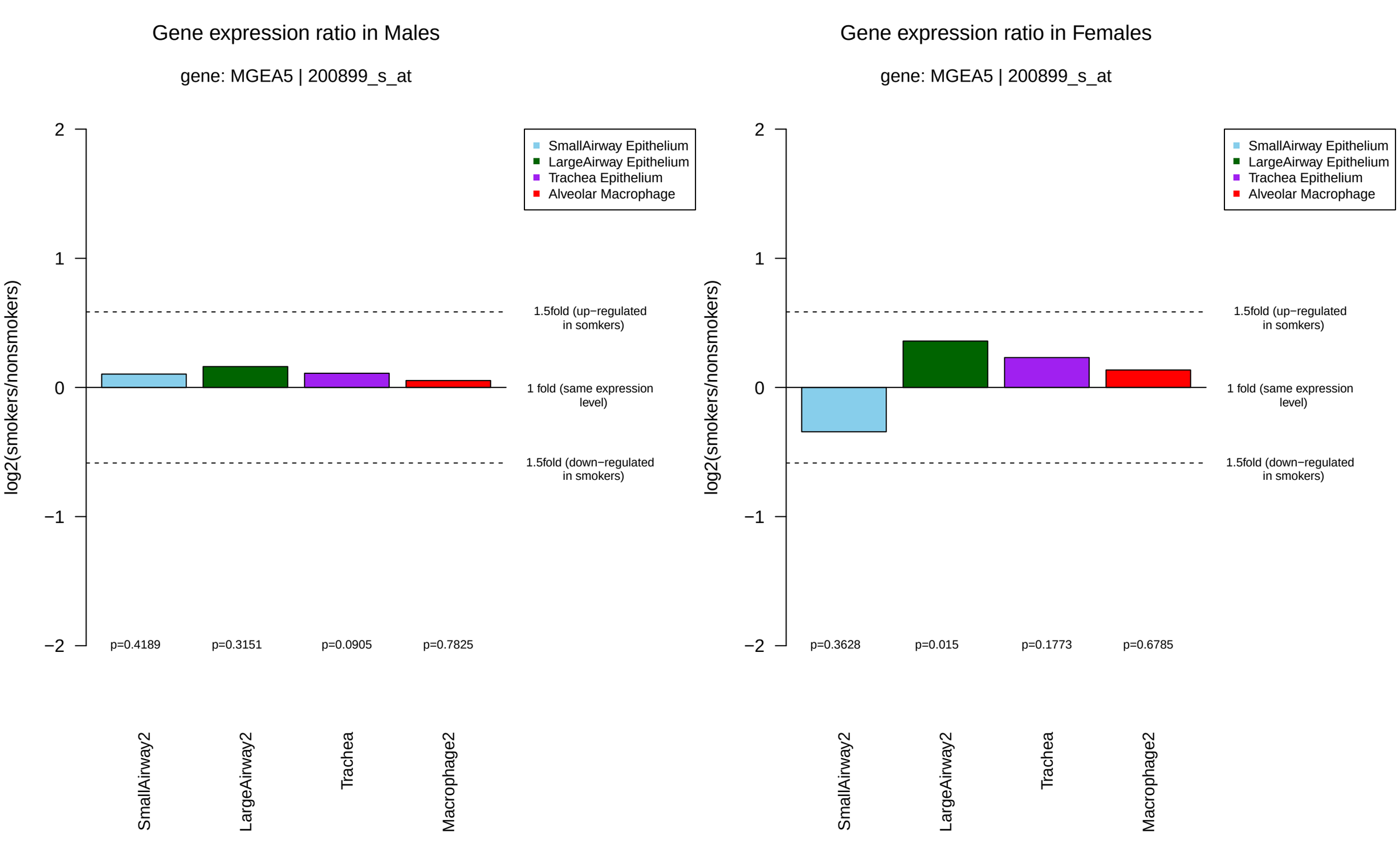
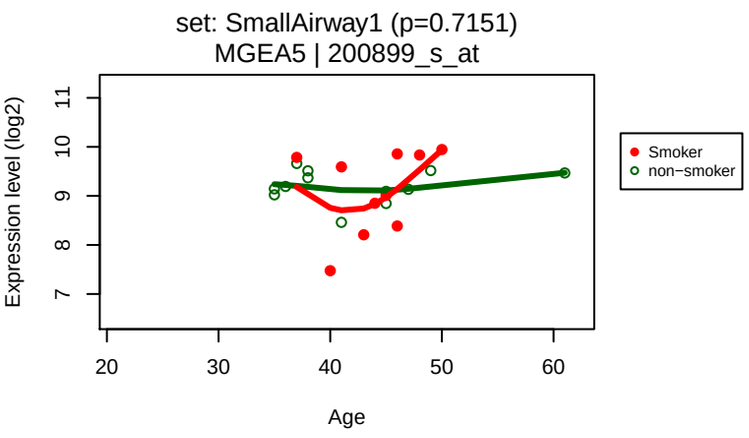
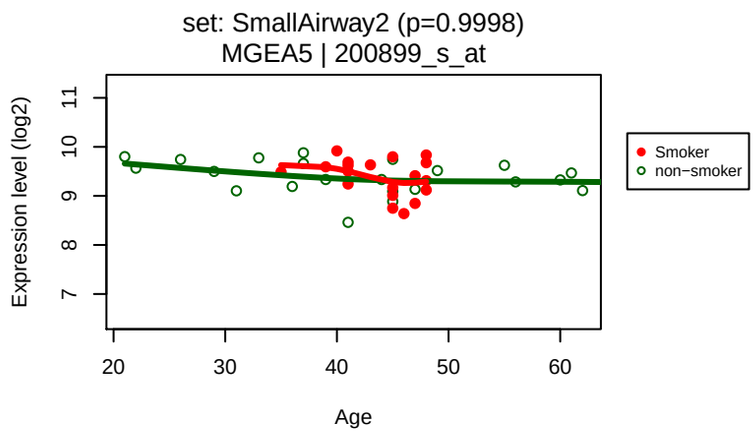
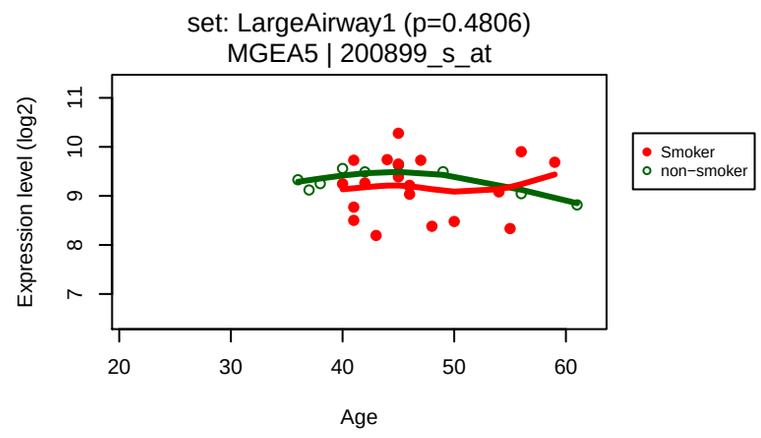

Gene expression of smokers and nonsmokers
gene: MGEA5 | 200899_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7151
p=0.9998
p=0.967
p=0.4806
p=0.0901
p=0.0154
p=0.0139
p=0.6572
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

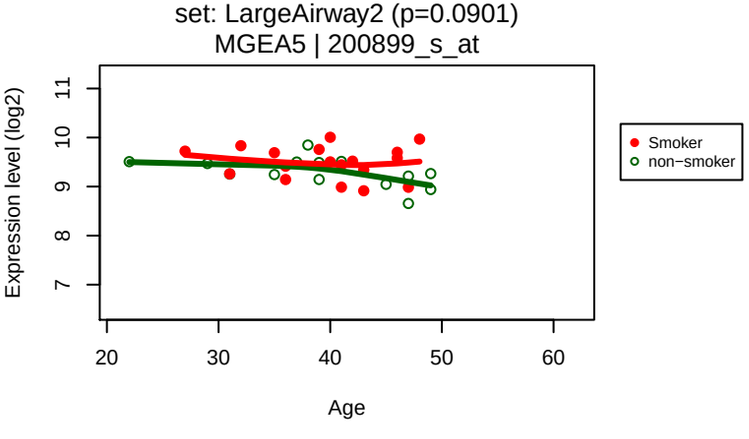

Gene expression ratio (smokers/nonsmokers)
gene: MGEA5 | 200899_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7151
p=0.9998
p=0.967
p=0.4806
p=0.0901
p=0.0154
p=0.0139
p=0.6572
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

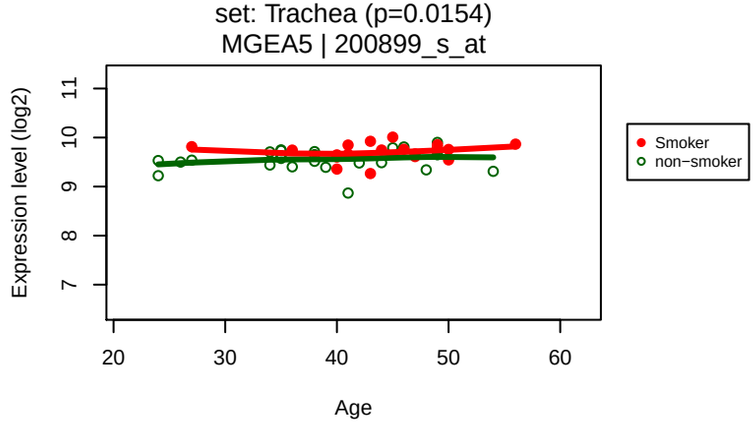

Gene expression ratio in Males
gene: MGEA5 | 200899_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4189
p=0.3151
p=0.0905
p=0.7825
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MGEA5 | 200899_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3628
p=0.015
p=0.1773
p=0.6785
SmallAirway2
LargeAirway2
Trachea
Macrophage2

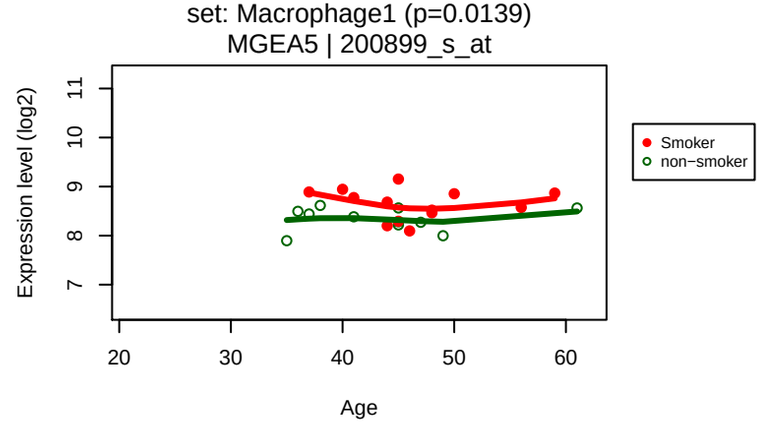

set: SmallAirway1 (p=0.7151)
MGEA5 | 200899_s_at
Expression level (log2)
Age
Smoker
non−smoker

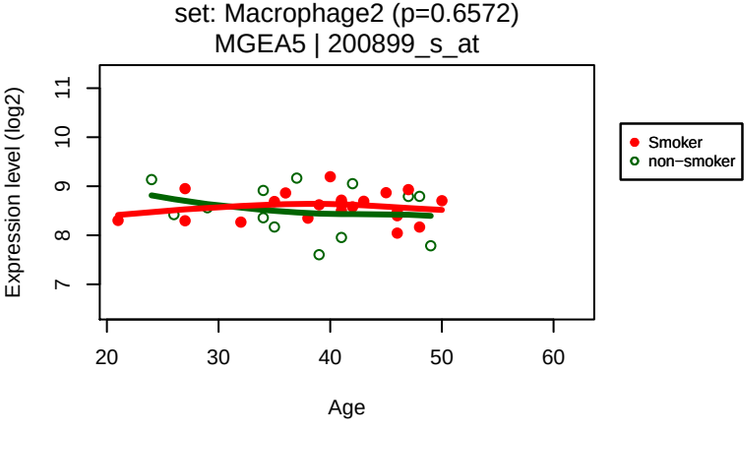

set: SmallAirway2 (p=0.9998)
MGEA5 | 200899_s_at
Expression level (log2)
Age
Smoker
non−smoker

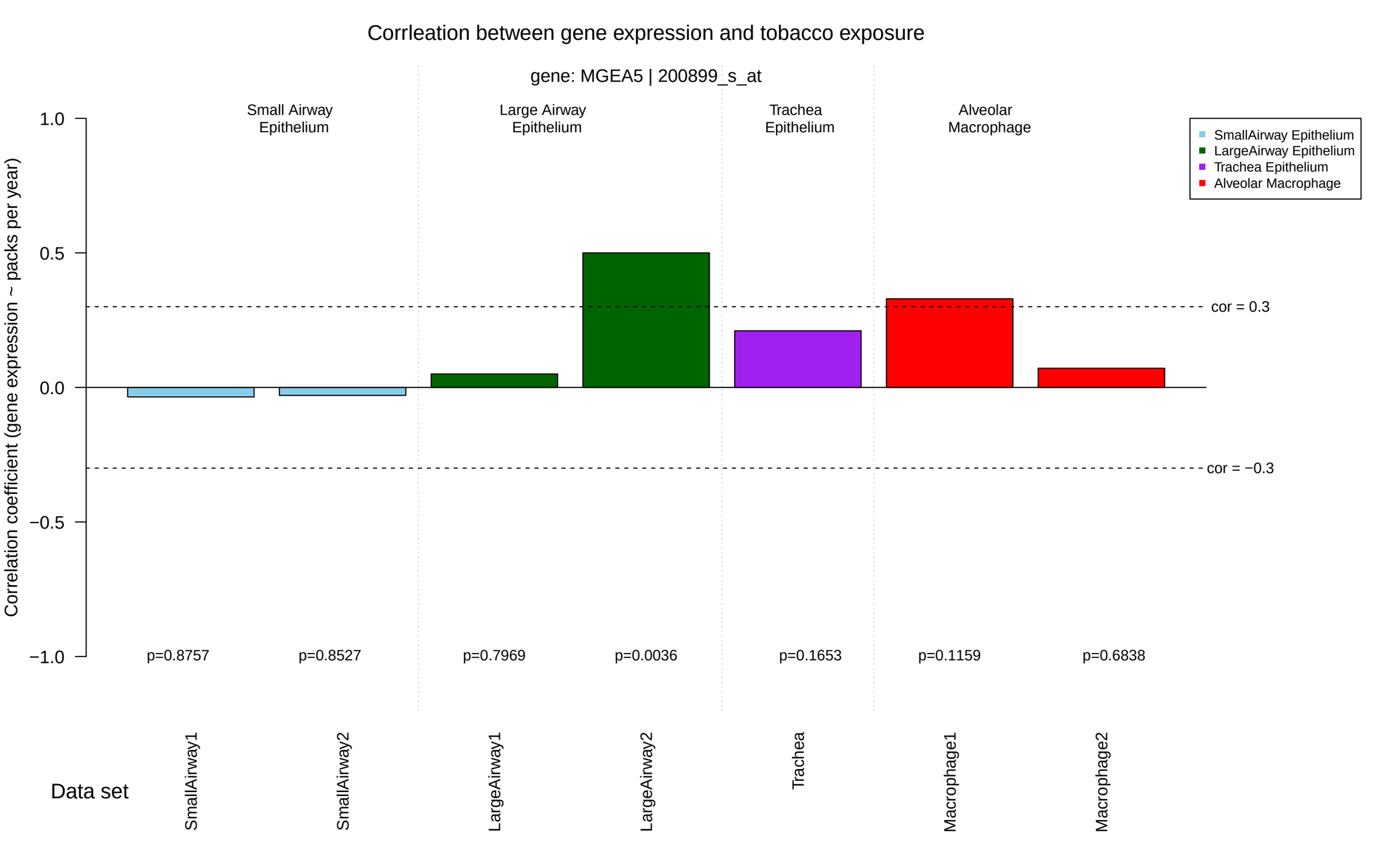

set: LargeAirway1 (p=0.4806)
MGEA5 | 200899_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0901)
MGEA5 | 200899_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0154)
MGEA5 | 200899_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0139)
MGEA5 | 200899_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6572)
MGEA5 | 200899_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MGEA5 | 200899_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8757
p=0.8527
p=0.7969
p=0.0036
p=0.1653
p=0.1159
p=0.6838
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2